

Appendix A: 3D Print Assembly and Detailed Drawing

Appendix A.1 3D Print Assembly

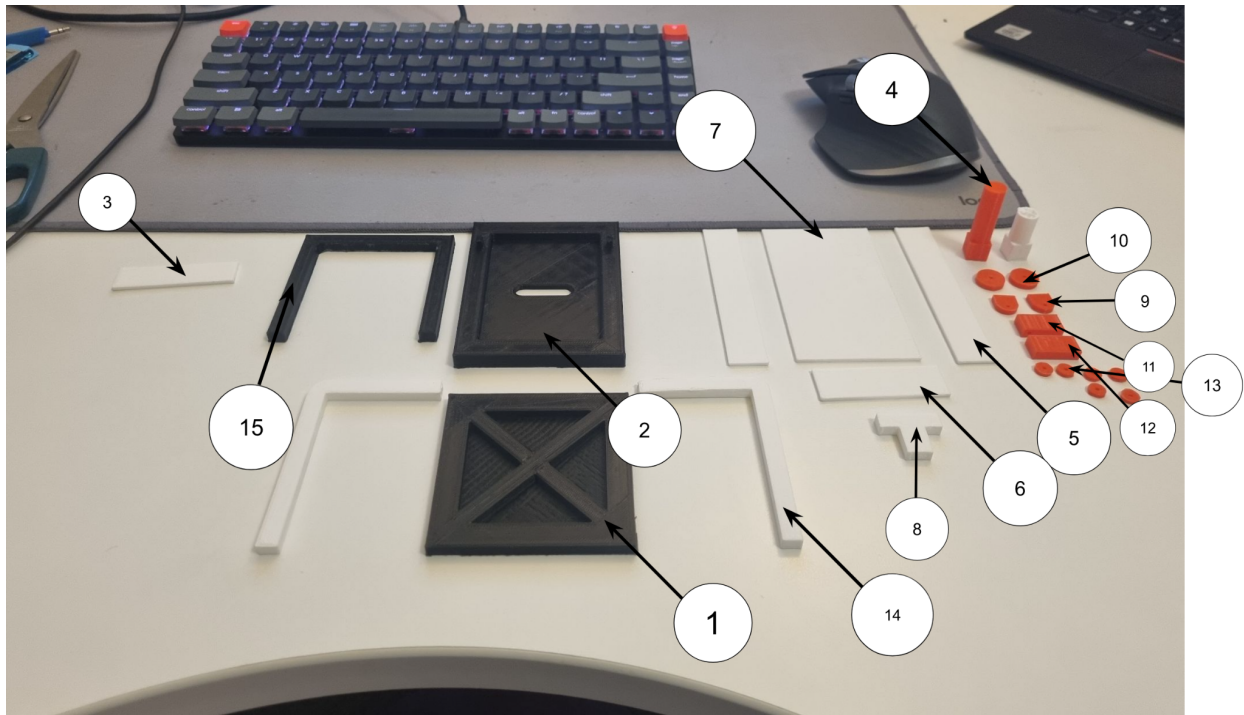


Figure 55: Parts for 3D assembly

Bill of Materials

Part Number	Part Name	Material	Quantity
1	Base Platform	Cast Alloy Steel	1
2	Backing Plate	1060 Alloy Aluminium	1
3	Motor Mounting Platform	1060 Alloy Aluminium	1
4	Motor	Siemens configured motor with gearbox	2
5	Cover Panel	ABS	2
6	Top Cover Panel	ABS	1
7	Back Cover Panel	ABS	1

8	Bearing support	1060 Alloy Aluminium	1
9	Bearing holder	1060 Alloy Aluminium	2
10	Pinion	4140 Steel Alloy	2
11	Top Guide Wheel Arm	1060 Alloy Aluminium	1
12	Bottom Guide Wheel Arm	1060 Alloy Aluminium	1
13	Guide Wheels	Manufactured Guide Wheels	8
14	Side Safety Rail	ABS	2
15	Back Safety Rail	ABS	1

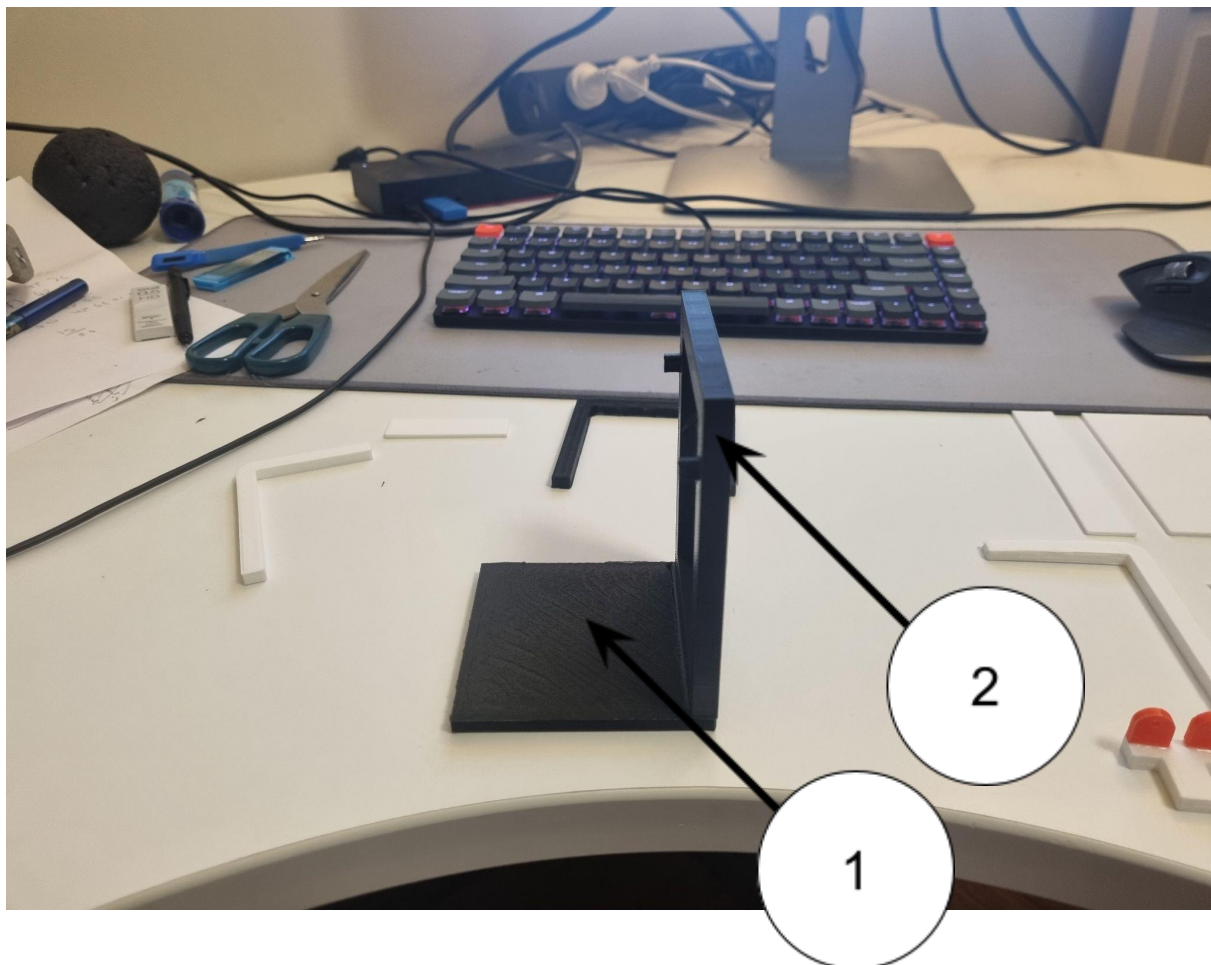


Figure 56: Assembly Step 1

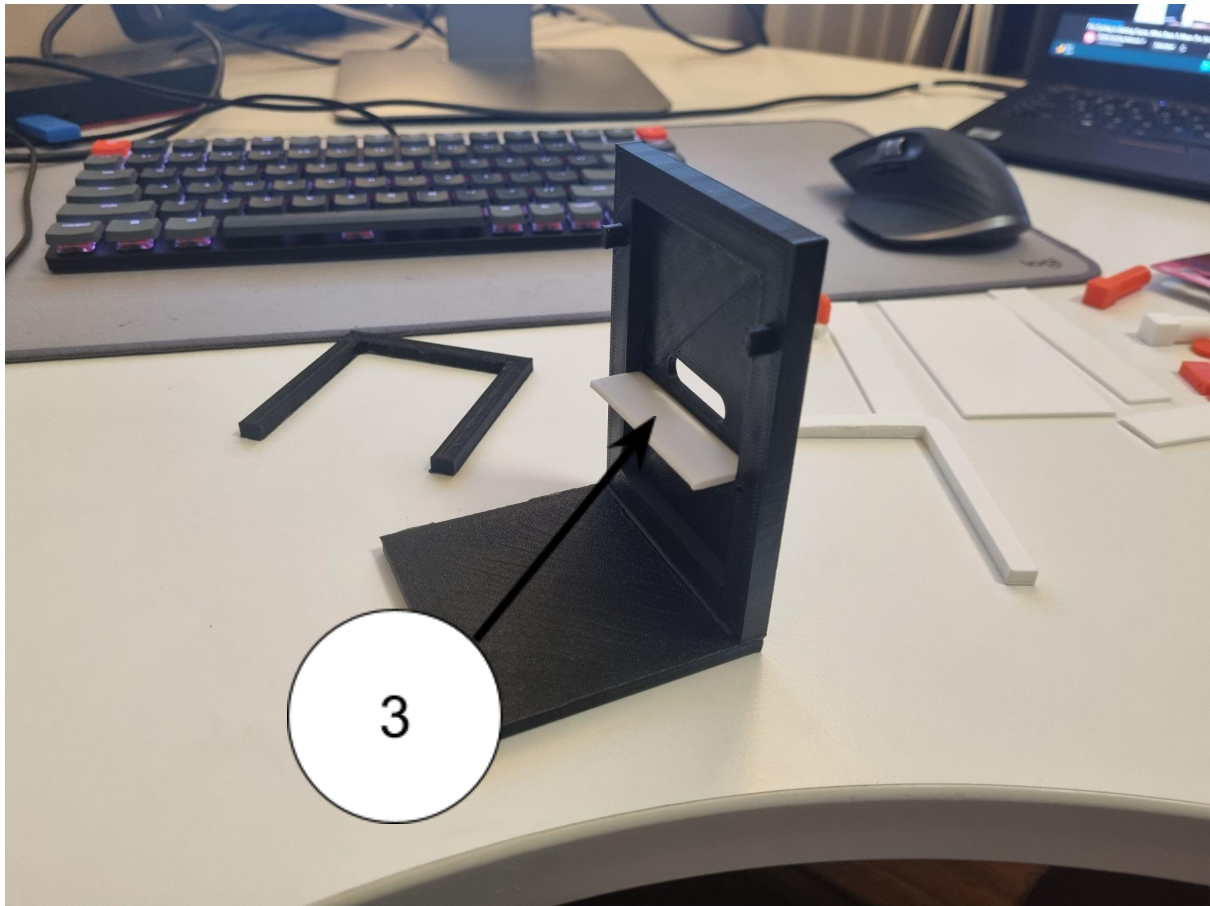


Figure 57: Assembly Step 2

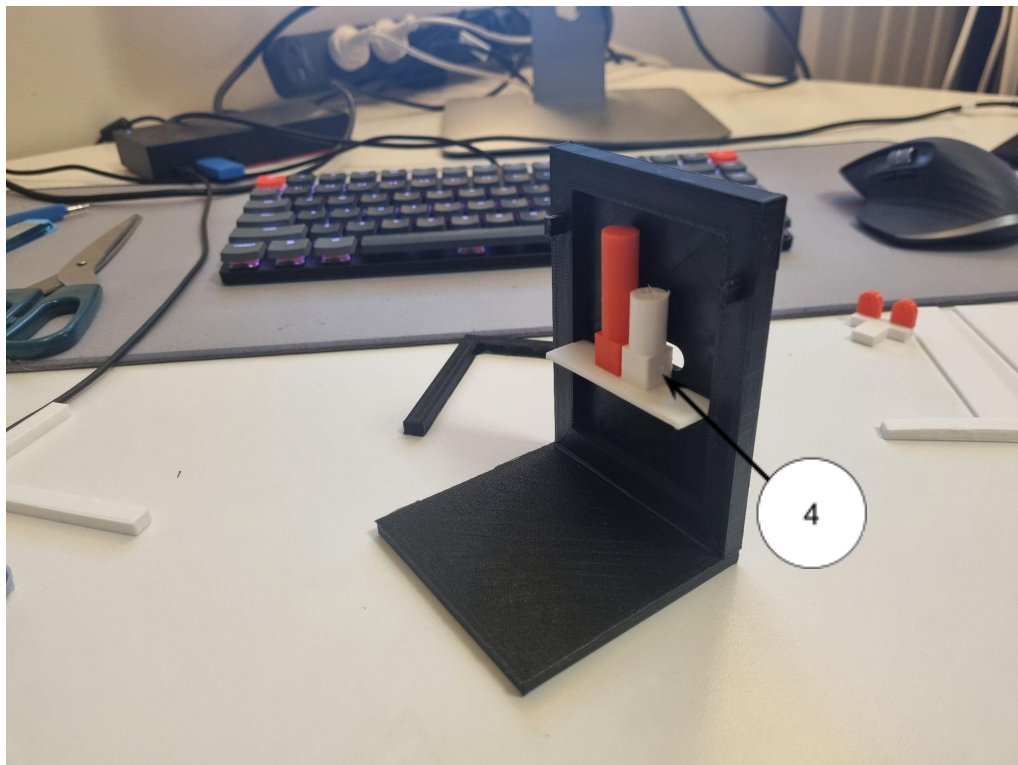


Figure 58: Assembly Step 3

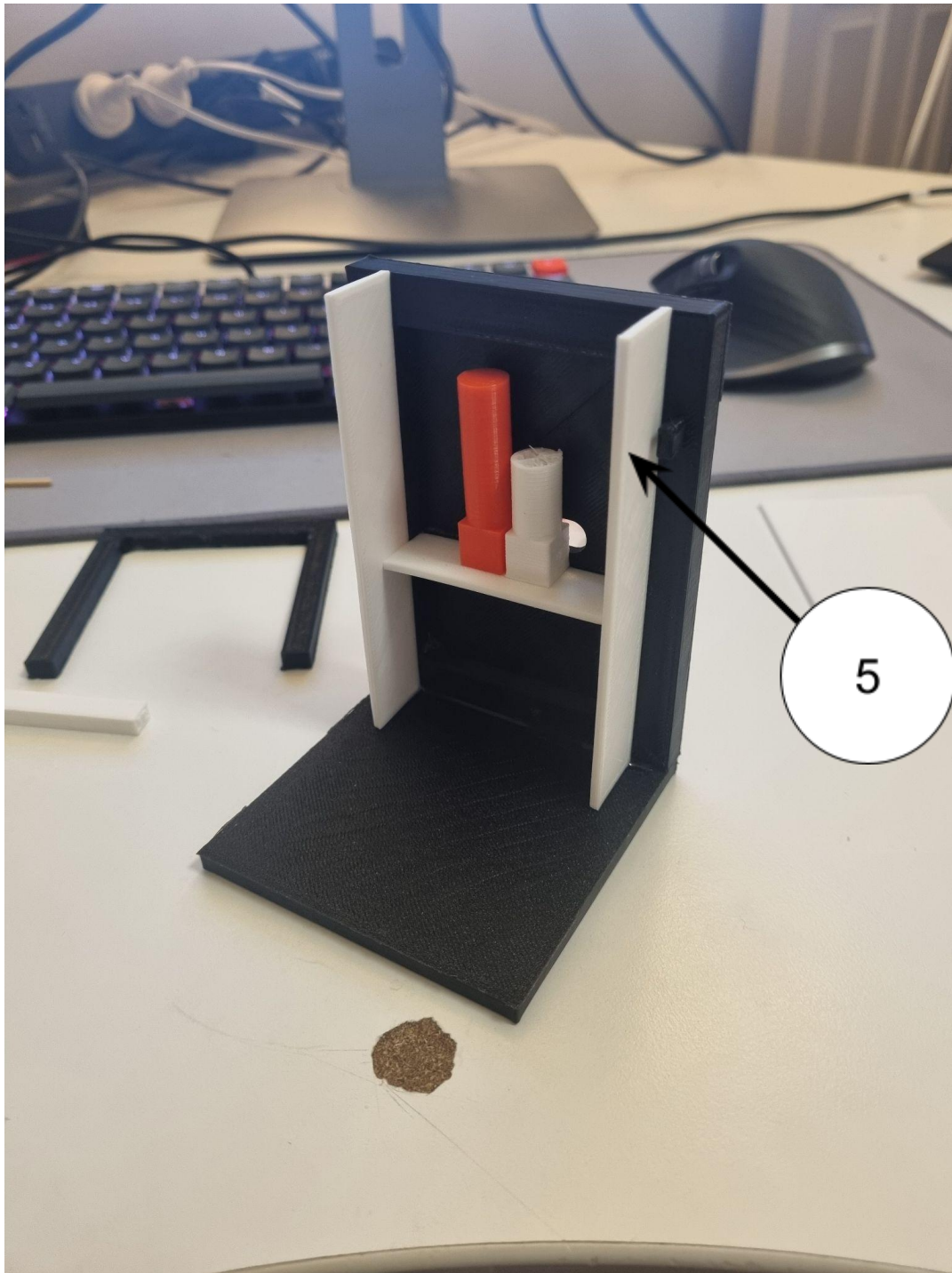


Figure 59: Assembly Step 4

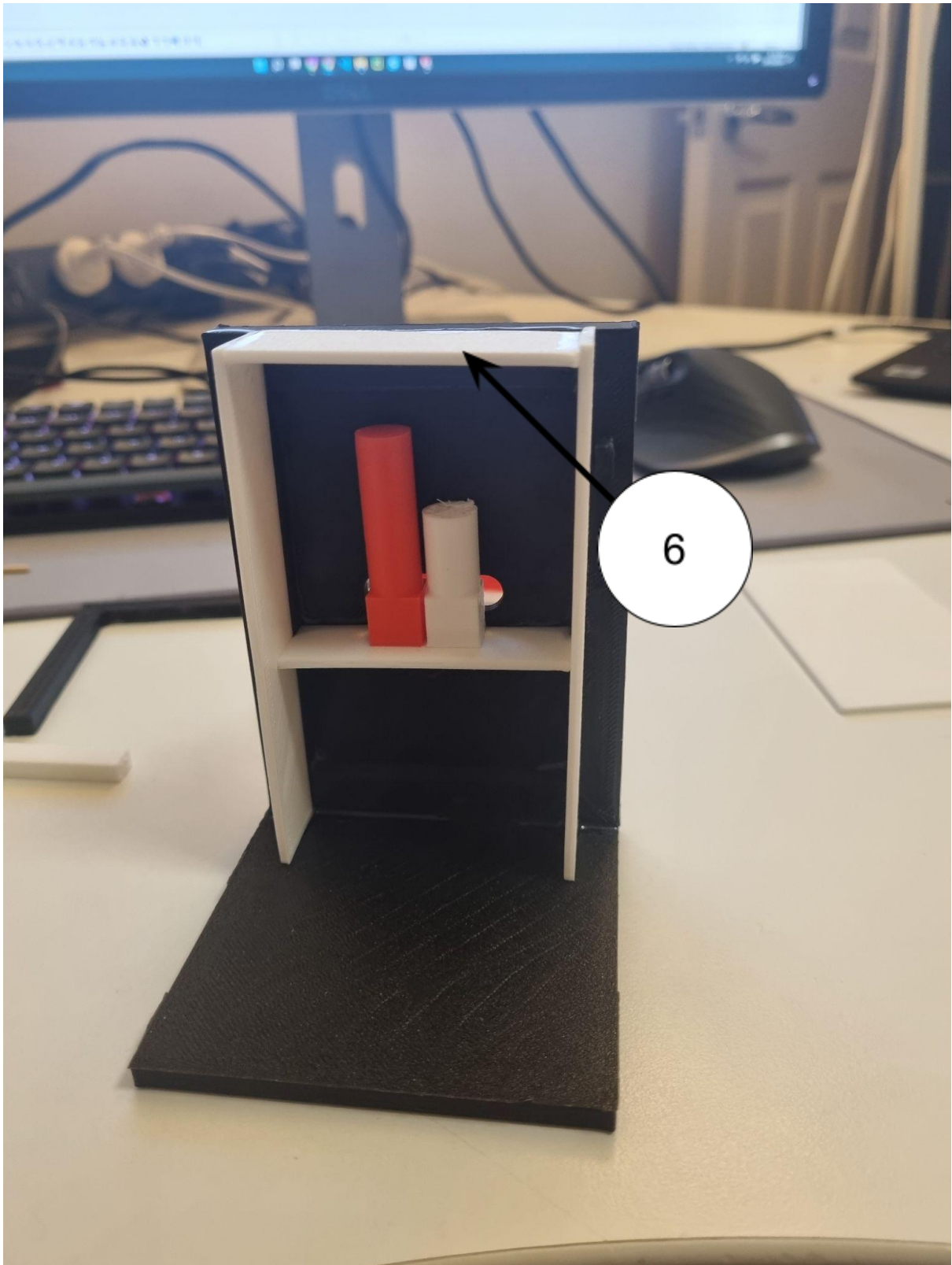


Figure 60: Assembly Step 5

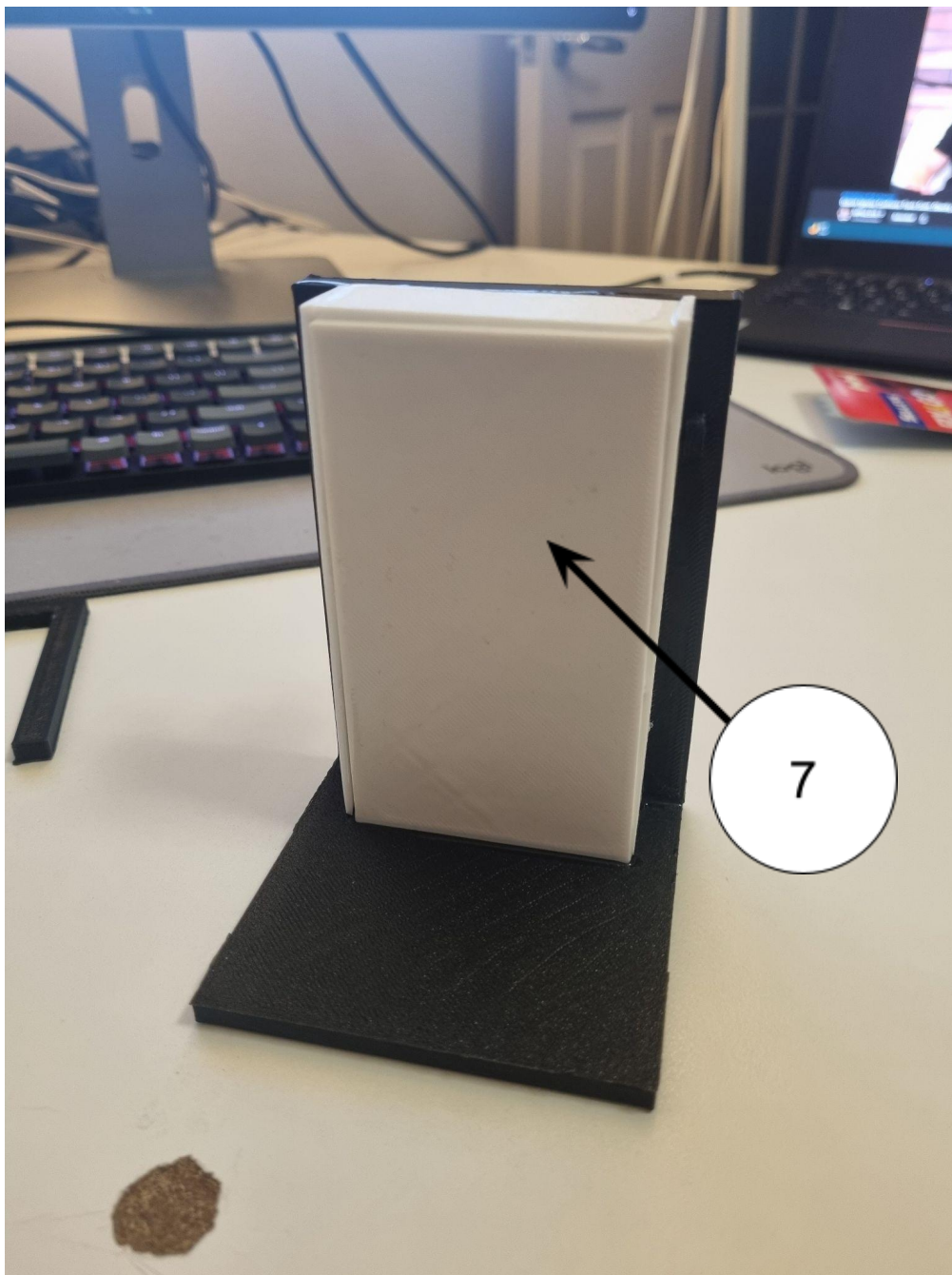


Figure 61: Assembly Step 6

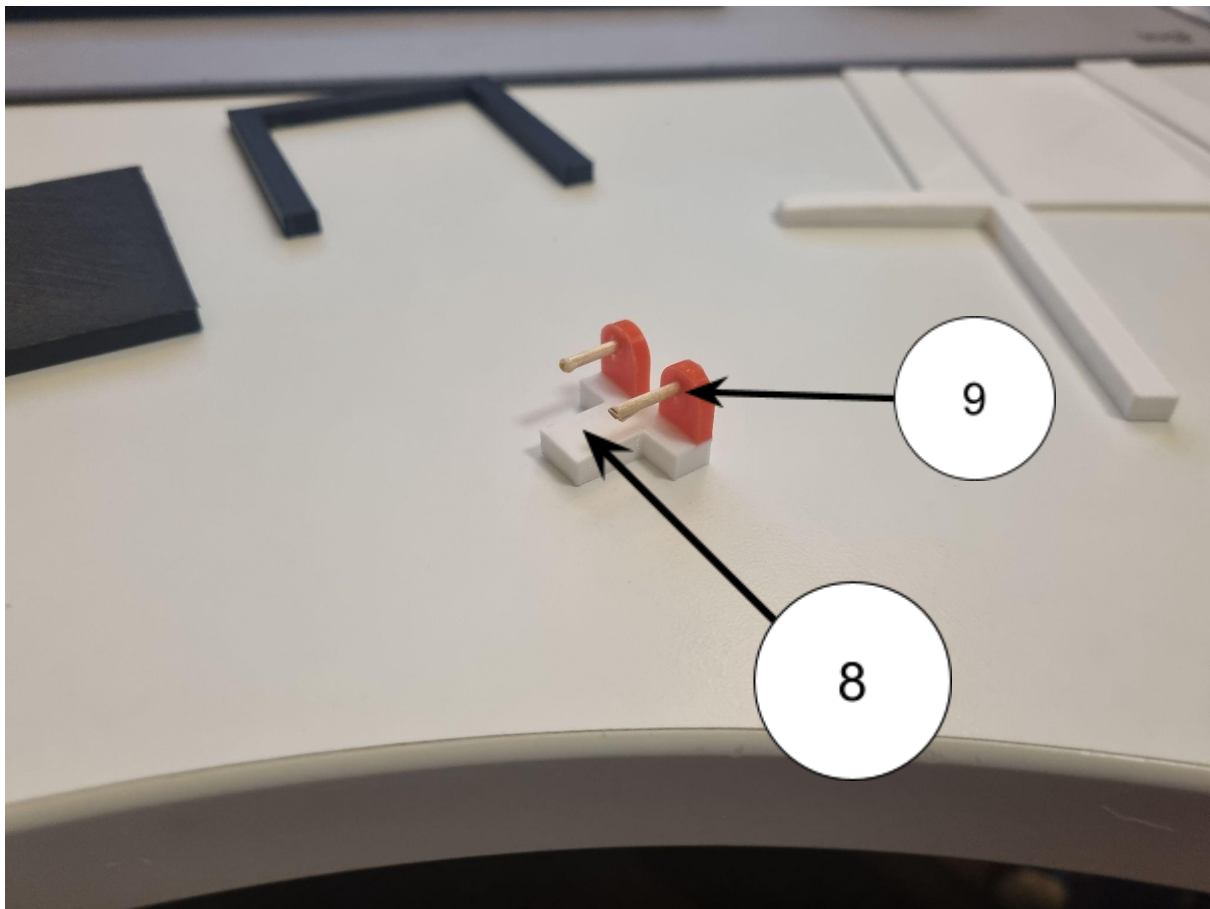


Figure 62: Assembly Step 7

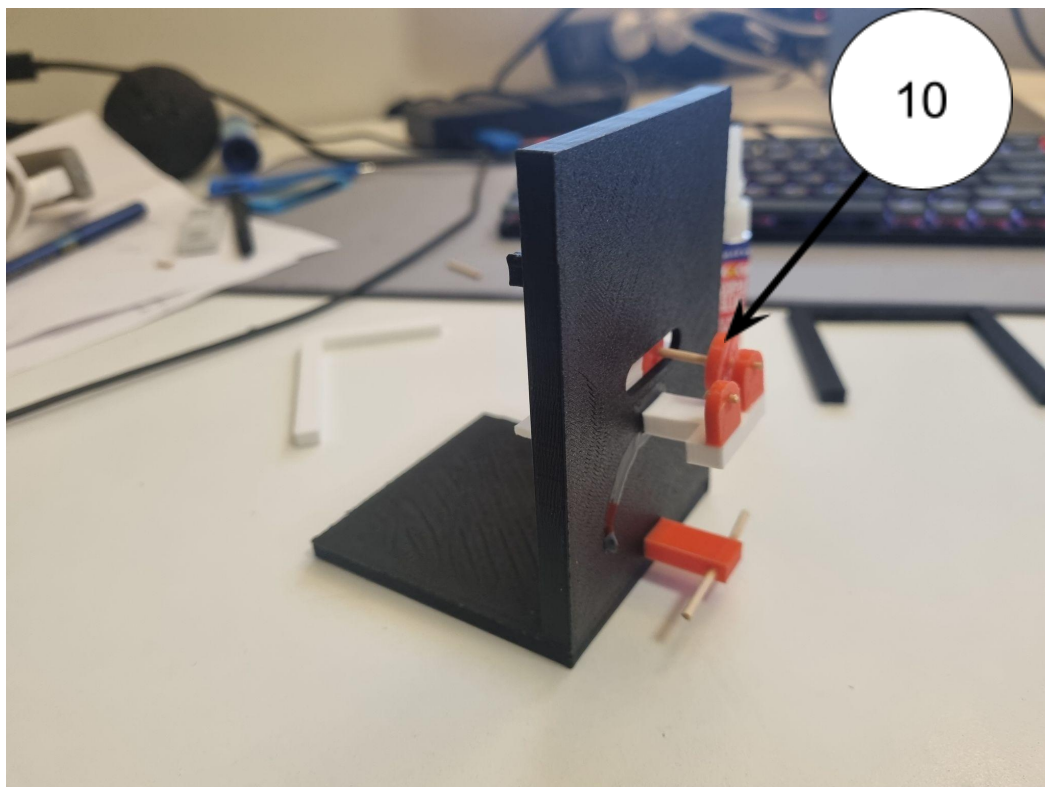


Figure 63: Assembly Step 8

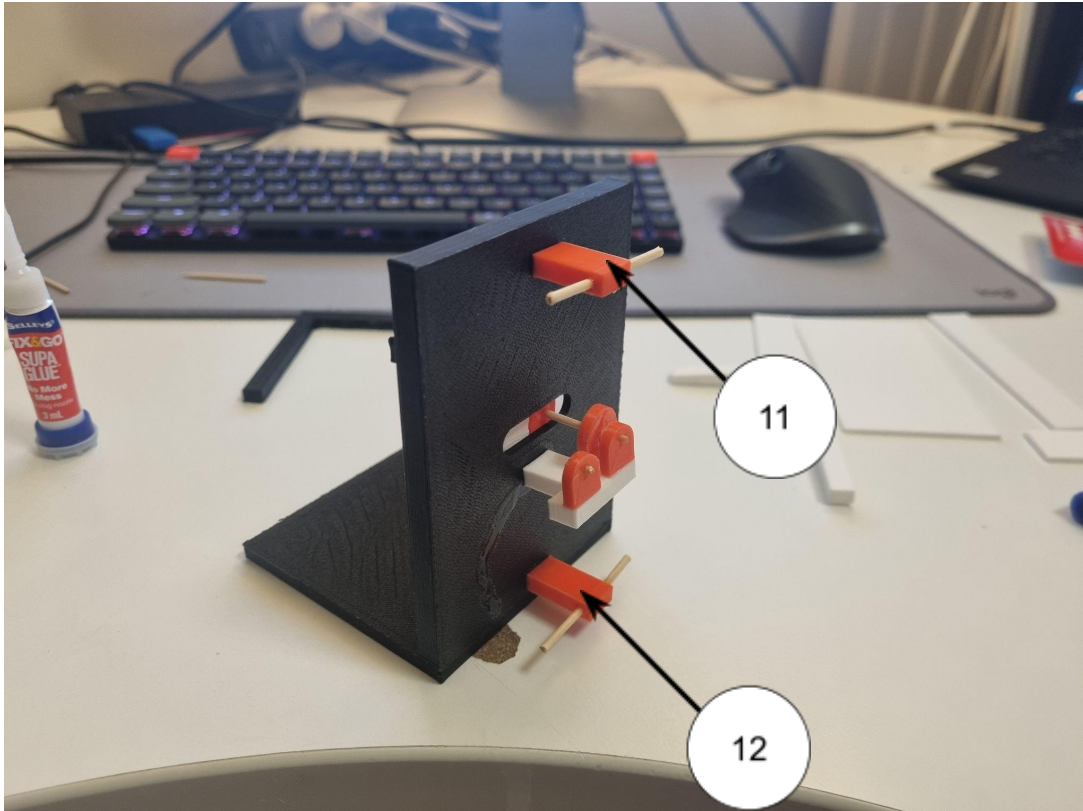


Figure 64: Assembly Step 9

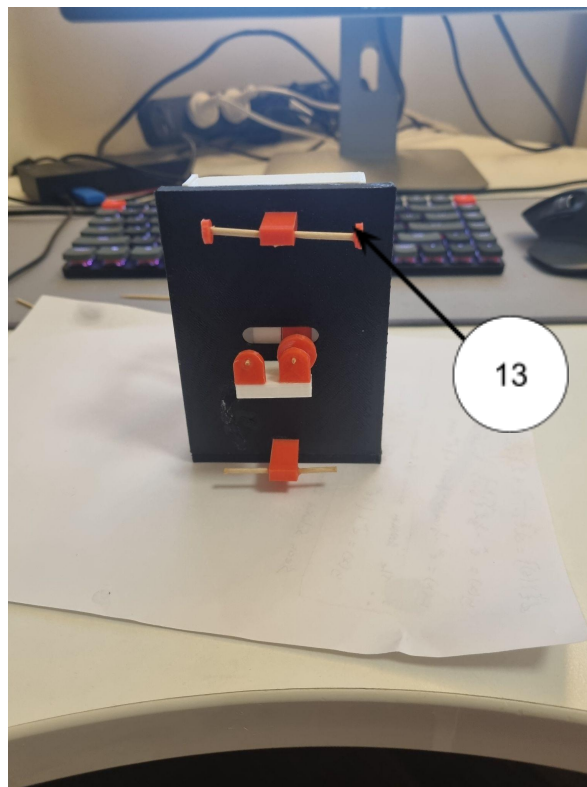


Figure 65: Assembly Step 10

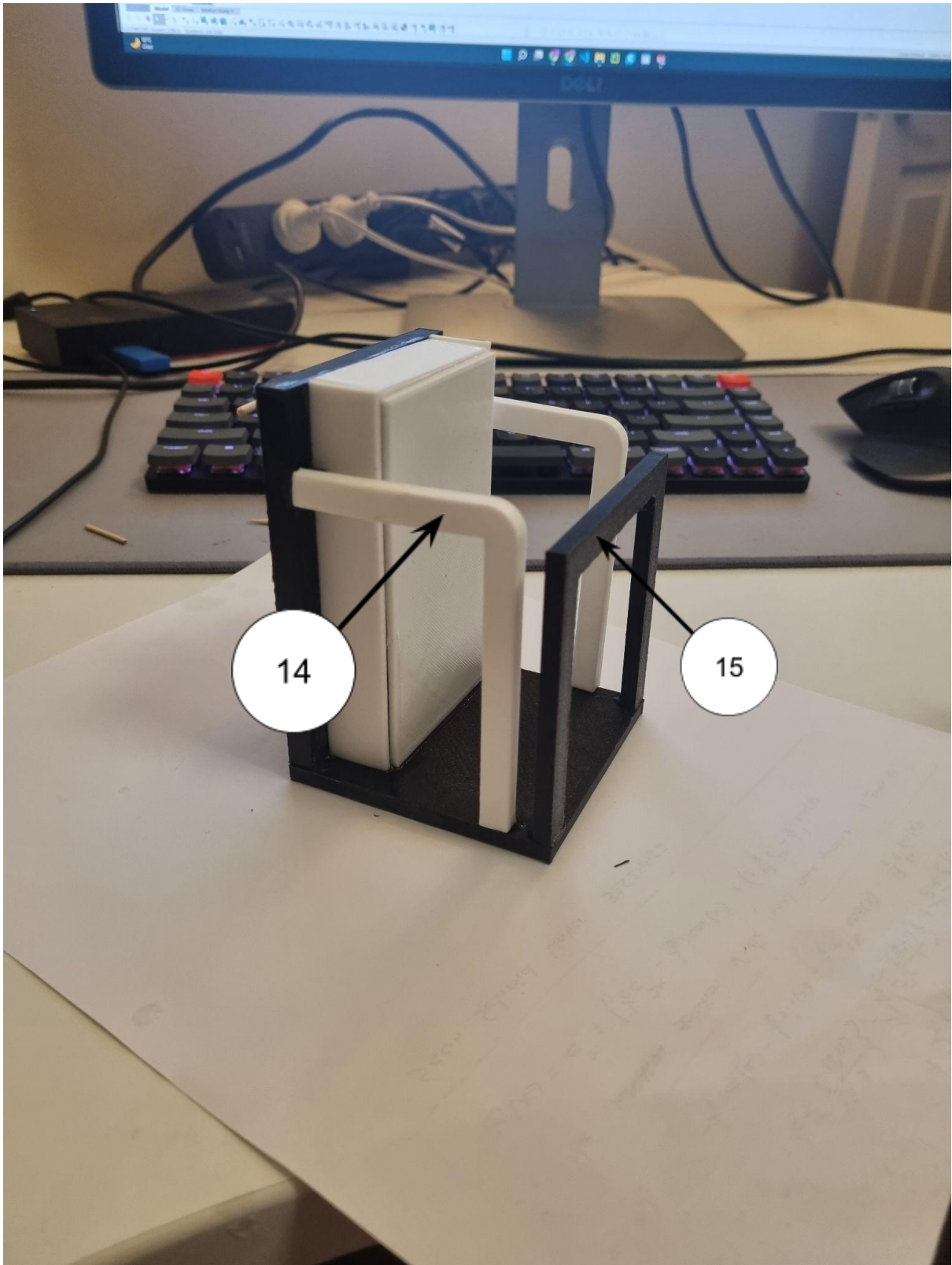


Figure 66: Assembly Step 11



Figure 67: Final Assembly

Appendix A.2 Detailed Drawing

